

Microbes in Human Welfare - 100 MCQs

NCERT Nichod : Biology Practice Series NEET 2025



Good evening
↓
available?
↓
Ready ⇒

Microbe in Human Welfare



BIOLOGYLIVE



NEET 2025 Countdown

Days

88

Hours

05

Minutes

59

Seconds

49

Every second counts! Keep pushing forward.

Physics

Chemistry

Biology

1 Physical World and Measurement

Theory

Lecture

Practice

Revised



Progress



Tests



Planner



Reports



Overall Performance

0.0%



Total Tests

0



Recent Tests

0

Last 30 days



Progress



Tests



Planner



Reports

Add Test Score

Test Type

Subject Test

Full Test

Subject

Physics

Chapter

Physical World and Measurement

Score

Max: 180

/ 180

Remarks (Optional)

Add any remarks about the test...

+ Add Test Score



Progress



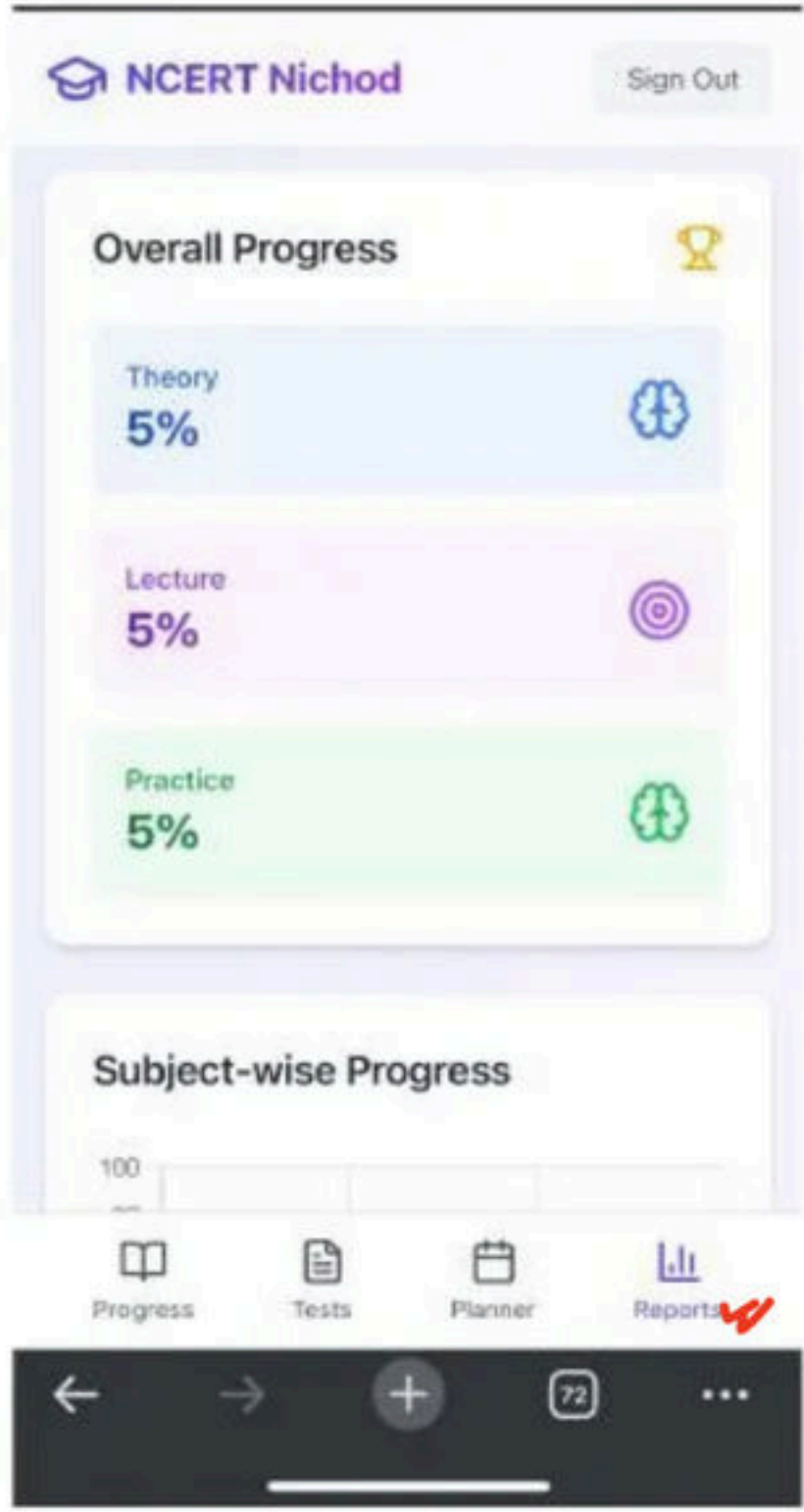
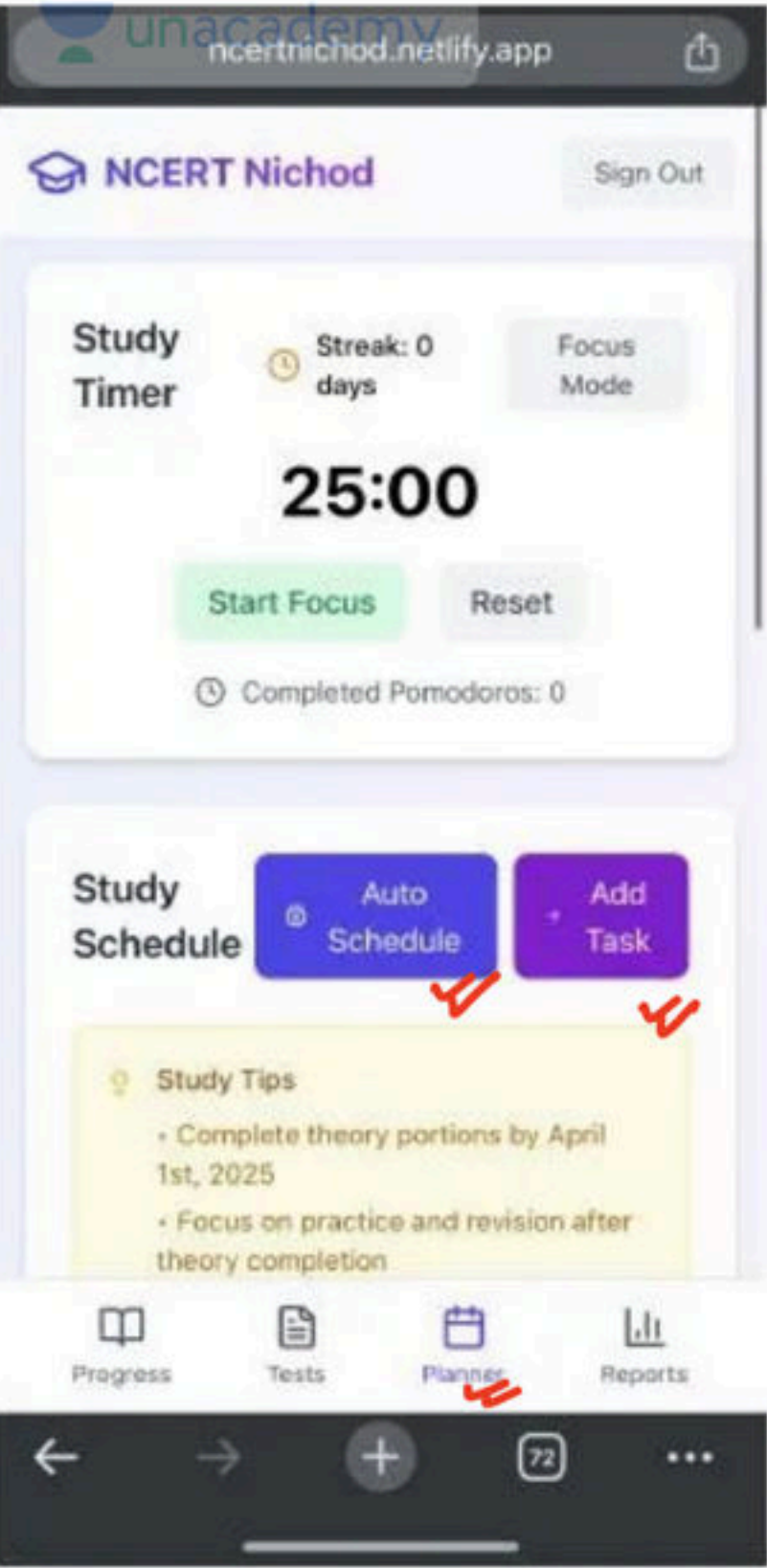
Tests



Planner



Reports



1. Methanogenic bacteria are not found in:

- (1) Rumen of cattle
- (2) Gobar gas plant
- (3) Bottom of water-logged paddy fields
- (4) Activated sludge.

✓ anaerobic bacteria



2. The vitamin whose content increases following the conversion of milk into curd by lactic acid bacteria is:

(1) Vitamin C

(2) Vitamin D

✓(3) Vitamin B₁₂

(4) Vitamin E.



3. The starter or inoculum is added to the fresh milk in order to convert milk into curd and improves its nutritional quality by increasing :-

✓ (1) Vitamin- B₁₂ ↩

(2) Vitamin-B

(3) Vitamin- B₂

(4) All of these



4. The lower the BOD of sample water, indicates that

(1) It is highly polluted

✓ (2) It is less polluted

(3) It is moderately polluted

(4) Pollution level cannot be determined



5. Which of the combinations is correct?

Column I

Column II

A Saccharomyces

1 Bacteria

B Trichoderma

2 Cyanobacteria

C Lactobacillus

3 Fungi

D Nostoc

4 Yeast

(1) A - 4; B - 2; C - 1; D - 3

(2) A - 3; B - 4; C - 1; D - 2

(3) A - 1; B - 2; C - 3; D - 4

✓ (4) A - 4; B - 3; C - 1; D - 2



6. Match the columns I and II.

Column I

A Azospirillum

B Rhizobium

C Oscillatoria

D Glomus

Column II

P BGA

Q Fungi

R Symbiotic bacteria

S Free living

(1) A - 4; B - 2; C - 1; D - 3

(3) A - 1; B - 2; C - 3; D - 4

(2) A - 3; B - 4; C - 1; D - 2

✓ (4) A - 4; B - 3; C - 1; D - 2



7. Microbe used for bio control of pest butterfly caterpillars is

- ✓ (1) *Bacillus thuringiensis*
- (2) *Bacillus thermolactis*
- (3) *Bacillus tequilensis*
- (4) *Bacillus thioparans*



8. Make correct pair

- A. Citric acid $\longrightarrow$ P. *Aspergillus niger*
B. Butyric acid $\longrightarrow$ Q. *Clostridium butyricum*
C. Proteases $\times$ R. *Candida lipolytica*
D. Lipase $\times$ S. *Aspergillus oryzae*

✓ (1) A - P, B - Q, C - S, D - R

(2) A - R, B - Q, C - P, D - Q

(3) A - P, B - Q, C - R, D - S

(4) A - Q, B - R, C - S, D - P



9. The nutritive medium for growing bacteria and many fungi in the laboratory is called

- (1) Culture media ✓
- (2) Broth
- (3) Nutritive medium ✓
- ✓ (4) Both 1 and 3



10. Symbiotic association is exhibited by

A. Mycorrhiza ✓✓

B. Rhizobium ✓✓

C. Few BGA ✓✓

D. Yeast ✗

(1) A, C, D

(2) A, B, C, D

(3) B, C, D

✓ (4) A, B, C



11. Microorganisms used for production of curd/ dahi are commonly called

(1) Leishmania Acid Bacteria(LAB)

✓(2) Lactic Acid Bacteria(LAB) ✓

(3) Acetic Acid bacteria(AAB)

(4) Lichen Acid Bacteria(LAB)



12. Match the columns I and II.

Column I

A Khadi and village industries commission

B Ministry of Environment and forests

C IPM

Column II

P Biocontrol

Q Sewage treatment

R Yamuna action plan

(1) A – R, B – P, C – Q

✓ (2) A – Q, B – R, C – P

(3) A – P, B – R, C – Q

(4) A – Q, B – P, C – R



13. Identify the incorrect statement with reference of
biocontrol agents.

(1) They do not show any negative impact on crop plants.

✓ (2) They help to ~~increase~~^{decrease} the use of synthetic pesticides X

(3) They are significant in treating ecologically sensitive area.

(4) They do not affect non-target pests.



14. Statement A : ^{Fungi} Pectinases are used in cleaning of oily stains ^{lipase} X
Statement B : Yeast is placed under slime moulds under protista X

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- ✓ (4) Both Statements are incorrect.



15. Which one of the following helps in absorption of phosphorus from soil by plants?

- ✓ (1) Glomus ←
- (2) Rhizobium
- (3) Frankia
- (4) Anabaena



16. Which one of the following has been commercialised
as blood-cholesterol lowering agent?

(1) Streptokinase

(2) Cyclosporin A

✓ (3) Statins

(4) Trypsin -A



17. Statement A : While working on staphylococci, Alexander Fleming observed that Penicillium notatum inhibits the growth of the bacteria.

Statement B : This inhibiting chemical was commercially extracted and its full potential was established by Ernest Chain and Howard Florey..

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

✓(3) Both statements are correct.

(4) Both Statements are incorrect.



18. Roquefort cheese employs ____ for ripening.

- ✓ (1) Fungi ↪
- (2) Bacteria
- (3) Blue-green algae
- (4) Protist



19. Statement A : The [✓]municipal wastewater is also called sewage.

[✗]Statement B : In [✗]secondary treatment, useful aerobic microbes grow rapidly and form flocs.
→ Have Fungal filament with hairs

[✓](1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

(3) Both statements are correct.

(4) Both Statements are incorrect.



20. The Ladybird, and Dragonflies are useful to get rid of ____ and ____, respectively.

(1) Aphids, cotton ballworms

✓(2) Aphids, mosquitoes

(3) Mosquitoes, aphids

(4) Cotton ballworms, mosquitoes , respectively.



21. Select the correct answer ∴

i. Sewage is treated in STPs to make it less polluting. ✓

ii. STP stands for Sewage ~~test~~ plant ✗

treat.

✓ (1) Only i is correct

(2) Only ii is correct

(3) Both are correct

(4) None are correct



22 Identify the bacteria and fungi which was growing on in the culture plate which were plated by Alexander Fleming?

- ✓✓ (1) Staphylococci, Penicillium notatum
- (2) Streptococci, Penicillium notatum
- (3) Staphylococci, Saccharomyces cerevisiae
- (4) Streptococci, Saccharomyces cerevisiae



23. Study the following table :-

Table –1

i. <i>Aspergillus niger</i>	<u>A fungus</u>	<u>Citric acid</u> ✓
ii. <i>Saccharomyces cerevisiae</i>	Algae ✗	Ethanol
iii. <i>Lactobacillus delbruchii</i>	Bacterium	Pencillin ✗
iv. <i>Streptococcus</i> <i>sps</i>	<u>Bacterium</u>	<u>Clot buster</u>

The correct combination is :-

- (1) i & ii
(2) i & iv ✓
(3) i & iii
(4) iii & iv



24. A nitrogen fixing microbe associated with Azolla in rice fields is :- Anabaena

- (1) Frankia
- (2) Rhizobium
- (3) Spriulina
- (4) Anabaena



25. In sewage treatment, secondary treatment is considered highly significant, because ; BOD ↓

(1) It helps to remove debris from the sewage

✓ (2) It reduces the BOD level of sewage ✓

(3) It helps in the production of biogas

(4) It increases the organic content of sewage



26. Which of the following statements is correct?

- ✓ I. Single cell protein is rich in high quality proteins but poor in fat. ✓
- II. BOD is the amount of oxygen that would be consumed if all the inorganic matter in one litre of water were oxidised by bacteria. ✗
- ✓ III. Trichoderma sp are free living fungi that are effective biocontrol agents of several pathogens.
- ✓ IV. Baculoviruses are pathogens that attack insects and other arthropods.
- ✓ V. Bacillus thuringiensis is first used biopesticide.
- VI. Lactase enzyme is used in alcoholic fermentation. ✗
- ✓ VII. Mycorrhiza is a symbiotic association between fungi e.g. Glomus and vascular plant.

(1) II, III, VI, VII

✓ (2) I, III, IV, V, VII

(3) I, IV, VI

(4) II, VI





27. Biofertilizers include

- ✓ (1) Blue green algae, Rhizobia and other N₂ fixing bacteria
- (2) Blue green algae, Rhizobia and lichens
- (3) Rhizobia, other N₂ fixing bacteria and Non mycorrhizal fungi
- (4) Blue green algae, Rhizobia and mycorrhizal fungi



28. Which one of the following alcoholic drinks is produced without distillation?

v. imp

- ✓ (1) Wine →
- (2) Whisky
- (3) Rum
- (4) Brandy



29. The primitive prokaryotes responsible for the production of biogas from the dung of ruminant animals includes

✓✓ (1) Methanogens

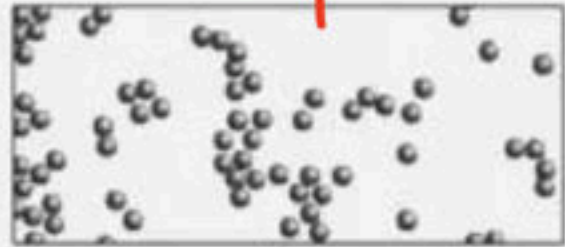
(2) Bacteria

(3) Pseudomonos

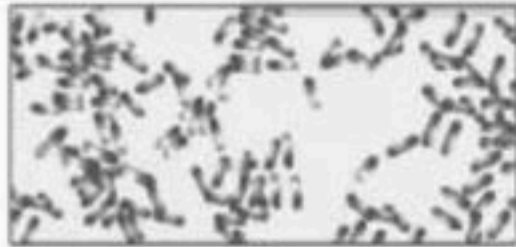
(4) Eubacteria



30. Identify the given bacteria based on its shape



A



B

- (1) A - Rod shaped - Bacilli, B - Spherical shaped - Cocci
- (2) A - Rod shaped - Cocci, B - Spherical shaped - Bacilli
- (3) A - Comma shaped - Vibrio, B - Spiral shaped - Spirilla
- ✓ (4) A - Spherical shaped – Cocci B - Rod shaped - Bacilli ✓



31. The bottled juices are clarified by the use of enzyme

- ✓ (1) Pectinases
- (2) Lipases
- (3) Amylases
- (4) Lactases



32. Methanogens are _____ and they produce methane by _____ respiration.

- ✓ (1) Archaeobacteria, Anaerobic
- (2) Archaeobacteria, Aerobic
- (3) Cyanobacteria, Anaerobic
- (4) Cyanobacteria, Aerobic



33. Assertion : Some bacteria produce vitamin ✓

Reason : Vitamins are obtained when they are alive ✗ bacterial fermentation process.

(1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.

(2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.

✓ (3) Assertion is correct but Reason is incorrect.

(4) Both Assertion and Reason are incorrect



34. In the primary treatment of sewage, the soil and small pebbles are removed by :- *Sedimentation*

(1) Filtration

(2) Condensation

✓ (3) Sedimentation

(4) Distillation



STP

35. Match the Column I with Column II.

Column I		Column II	
A	The stage in which physical treatment of sewage is done	P	Anaerobic digestion of activated sludge and production of biogas
B	The stage in which biological treatment of sewage is done	Q	Activated sludge
C	The names of sediment and supernatant in primary treatment	R	Aeration tanks
D	___ is carried to aeration tanks from primary settling	S	Primary effluent
E	Site of flocs growth	T	Primary sludge, effluent
F	Name of the sediment of secondary settling	U	Secondary treatment
G	The function of sludge digester	V	Primary treatment

- (1) A – P, B – R, C – T, D – V, E – Q, F – S, G – U
 (2) A – V, B – U, C – T, D – S, E – R, F – Q, G – P
 (3) A – P, B – Q, C – R, D – S, E – T, F – U, G – V
 (4) A – V, B – U, C – P, D – Q, E – R, F – S, G – T



→ 36. Assertion : Nowadays people realised that there is a large pressure to switch to organic farming.

Reason : Biofertilizers are organisms that enrich the nutrient quality of soil.

✓ (1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.

(2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.

(3) Assertion is correct but Reason is incorrect.

(4) Both Assertion and Reason are incorrect



37. Ernest chain and Howard Florey's contribution was: -

- (1) Discovery of Streptokinase
- (2) Discovery of DNA sequence
- ✓ (3) Establishing the potential of penicillin as an effective
antibiotic ✓
- (4) Production of genetically engineered insulin



38. Select the odd one w.r.to biofertilisers

(1) Bacteria

(2) Fungi

(3) BGA

✓✓ (4) viruses



39. Statement A : Organ transplantation patients are given immunosuppressive drugs.

Statement B : Immunosuppressive drug used is Cyclosporin -A

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

✓(3) Both statements are correct.

(4) Both Statements are incorrect.



40. Match the items in Column I and Column II then choose correct order.

Column I		Column II	
A	Blue green algae as biofertilizer	P	Ecto mycorrhiza
B	Fungi as biofertilizer	Q	<i>Trichoderma</i>
C	Bacteria as biofertilizer	R	<i>Rhizobium</i>
D	Fungi as Biocontrol	S	<i>Anabaena</i>

- ✓ (1) A - S, B - P, C - R, D - Q
(2) A - P, B - Q, C - S, D - R
(3) A - P, B - Q, C - R, D - S
(4) A - P, B - R, C - P, D - Q



41) The technology of biogas production in India was developed by the collaboration of ____ and ____.

(1) A - Indian Agricultural Research Commission, B - Khadi and Village Industries Commission

✓ (2) A - Indian Agricultural Research Institute, B - Khadi and Village Industries Commission

(3) A - National College for Agriculture and Development, B - Indian Agricultural Research Institute

(4) A - National College for Agriculture and Development, B - Khadi and Village Industries Commission



42. In STPs after reduction of the BOD, the remaining major part of sludge is pumped into large tanks called

- ✓✓ (1) Anaerobic sludge digesters ✓✓
- (2) Aerobic sludge digesters
- (3) Biological sludge digesters
- (4) None of these



43. Which one (s) is/are correct about Cyclosporin A?

(1) It is produced by the fungus Trichoderma polysporum ✓

(2) It is used as an immuno suppressive agent during organ transplantation

✓ (3) Both (1) and (2)

(4) It is produced by a type of yeast, *Monascus purpureus*



44. Secondary treatment of sewage

- (1) Removes grit and large pieces of organic matter
- (2) Involves shredding, churning, filtration and sedimentation
- (3) Does not require aeration
- ✓ (4) Involves microbial digestion of organic matter



45. Assertion : Nowadays production/use of herbicide resistant GM crops has increased.

Reason : It eliminates weeds from the field without the use of manual labour.

(1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.

☒ (2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.

(3) Assertion is correct but Reason is incorrect.

(4) Both Assertion and Reason are incorrect



46. Match the following organisms with the products they produce
Select the correct option.

Column I		Column II	
A	<i>Lactobacillus</i>	P	Cheese
B	<i>Saccharomyces cerevisiae</i>	Q	Curd
C	<i>Aspergillus niger</i>	R	Citric acid
D	<i>Acetobacter aceti</i>	S	Bread
		T	Acetic acid

(1) A - Q; B - S; C - T; D - R

(3) A - R; B - S; C - T; D - P

✓ (2) A - Q; B - S; C - R; D - T

(4) A - Q; B - P; C - R; D - T



47 Statement A : A person admitted to hospital as he had myocardial infarction. A doctor injected him 'Streptokinase'.

Statement B : 'Streptokinase' is a clot buster

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

✓ (3) Both statements are correct.

(4) Both Statements are incorrect.



48. Select incorrect w.r.t sewage treatment plants. :-

(1) In anaerobic sludge digester, gases like methane, H_2S

And CO_2 are also produced.

✓ (2) Bacterial flocs are allowed to sediment in setting tank
when BOD of sewage is high. ✗

(3) Primary treatment of sludge is based on sequential
filtration and sedimentation.

(4) Activated sludge is pumped into anaerobic sludge
digesters to digest the bacteria and fungi



49. Which one of the following is used as an immunosuppressive agent in organ-transplant patients?

- ☒ (1) Cyclosporin A
- (2) Statins
- (3) Antiseptics
- (4) Analgesics



50. Streptokinase produced by the bacterium ___ A ___ and modified by genetic engineering is used as a ___ B ___ for removing clots from the blood vessels of patients who have undergone ___ C ___.

(1) A - Streptococcus, B - 'clot buster, C - hypotension

✓(2) A - Streptococcus, B - 'clot buster, C – myocardial infarction

(3) A - Streptococcus, B - 'clot booster, C – myocardial infarction

(4) A - Stapylococci, B - 'clot buster, C – myocardial infarction



51. Match the following

Column I

A Cyclosporin- A

B Streptokinase

C Statins

D Penicillin

Column II

P Clot busters

Q Antibiotic

R Immuno suppressive agent

S Blood cholesterol lowering agent

(1) A - S, B - P, C - R, D - Q

(3) A - P, B - Q, C - R, D - S

(2) A - R, B - S, C - P, D - Q

(4) A - P, B - Q, C - S, D - R

No option
matches



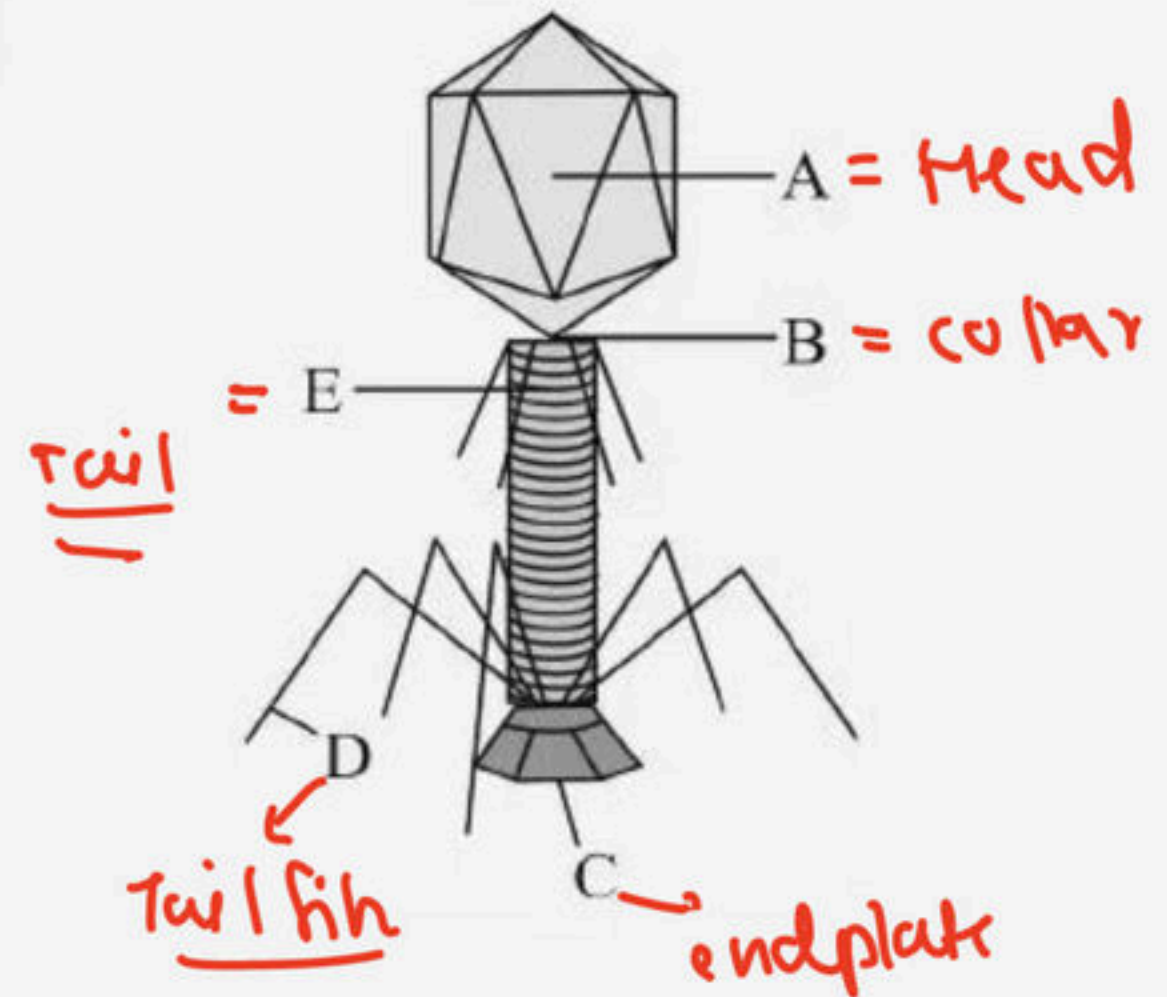
52. Which one of the following useful organism is not a part in production of curd/yoghurt?

- (1) Lactococci
- (2) Lactobacillus bulgaricus
- ☒ (3) Acetobacter aceti ✓
- (4) Lactobacillus acidophilus



53. Identify the different parts of Bacteriophage represented by alphabets (A-E):

- ✓ (1) Head, Collar, End plate, Tail fibre, Tail. ✓
- (2) Head, Collar, End plate, Tail, Tail fibre
- (3) Collar, Loop, End plate, Tail, Tail fibre
- (4) Head, Collar, Prong, Tail, Tail fibre



54. Select the correct answer

- i. A lot of cellulosic material present in the food of cattle is also present in the abomasum. rumin 10-15
- ii. The biogas plant consists of a concrete tank (15-25 feet deep) in which bio-wastes are collected.

- (1) Only i is correct
- (2) Only ii is correct
- (3) Both are correct
- ☒ (4) None are correct



55. Ruminant animals can digest cellulose in their food, where as human beings are unable to do so. This is because

- (1) Methanogens are present in human gut.
- (2) Cellulose is a complex sugar
- (3) Cellulose is a complex sugar
- ✓ (4) Methanogens are absent in human gut



56 What would happen if oxygen availability to activated sludge flocs is reduced?

- (1) It will slow down the rate of degradation of organic matter ✗
- ✓ (2) The center of flocs will become anoxic, which would cause death of bacteria and eventually breakage of flocs.
- (3) Flocs would increase in size as anaerobic bacteria would grow around flocs.
- (4) Protozoa would grow in large numbers.



57. Production of alcohol, antibiotics on an industrial scale, requires growing microbes in very large vessels called ____

- (1) Bioreactors
- ✓ (2) Fermentors
- (3) Settling tank
- (4) Culture plate



58. Treatment of wastewater is done by the :-

- (1) Photosynthetic blue green algae
- (2) Nitrogen fixing bacteria
- ✓ (3) Heterotrophic microbes
- (4) Chemosynthetic bacteria



59. STP treatment is carried out in two stages. They are

- (1) Biological treatment
- (2) Secondary treatment
- (3) Primary treatment →
- ✓ (4) All of them



60. Match the following list of bioactive substances and their roles:

Column I

A Statin

B Cyclosporin A

C Streptokinase

D Lipase

(1) A-Q, B-R, C-P, D-S

(3) A-S, B-P, C-Q, D-R

Column II

P Removal of oil stains

Q Removal of clots from blood vessels

R Lowering of blood cholesterol

S Immuno-suppressive agent

(2) A-S, B-Q, C-P, D-R

(4) A-R, B-S, C-Q, D-P



61. Identify the correct statement with regards to Tobacco

Mosaic Virus (TMV).

I. Rod-shaped

II. Spherical shaped

III. Visible after being magnified about 1, 00, 000 – 1, 50, 000X

(1) Only I is correct

(2) Only II, III is correct

(3) Only I, III is correct

(4) All are correct



62. The practice of pest control through biological or natural methods and using the chemical pesticides to the minimum is called ____

- (1) Integrated pest management
- (2) Integrated plant management
- (3) Intercropping
- (4) Organic farming



63. Secondary sewage treatment is mainly a

- (1) Biological process
- (2) Physical process
- (3) Mechanical process
- (4) Chemical process



64. The majority of Baculo viruses are used as biological control agents because

- (1) These are excellent candidates for species specific applications
- (2) They have no negative impact on non target plants and animals
- (3) Beneficial insects are being conserved
- (4) All the above



65. Which of the following microbial organisms can be used as biocontrol agents?

i. Lady bird beetle and NPV,

ii. Dragon flies,

iii. Bt and Gambusia fish,

iv. Trichoderma & Bt

(1) iii, iv

(2) i, ii, iii, iv

(3) i, iii, iv

(4) iv



66. Statins are produced by the ____

- (1) Yeast *Monascus purpureus*
- (2) Bacterium *Monascus purpureus*
- (3) Phycomycetes *Monascus purpureus*
- (4) Protist *Monascus purpureus*



67. The domestic sewage mixes with river water

- (1) Small animals like rats will die after drinking river water.
- (2) The increased microbial activity releases micronutrients such as iron.
- (3) The decreased microbial activity uses up dissolved oxygen.
- (4) The increased microbial activity uses up dissolved oxygen



68. Statement A : Baculoviruses are narrow spectrum insecticides.

Statement B : These viruses are excellent choices for biocontrol in an ecologically sensitive area for integrated pest management.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



69. Match the following list of bacteria and their commercially important products:

Choose the correct match:

Column I

A Aspergillus niger

B Acetobacter aceti

C Clostridium butylicum

D Lactobacillus

(1) A-Q, B-R, C-S, D-P

(3) A-S, B-R, C-Q, D-P

Column II

P Lactic acid

Q Butyric acid

R Acetic acid

S Citric acid

(2) A-Q, B-S, C-R, D-P

(4) A-S, B-P, C-R, D-Q



70. Which one of the following statement is incorrect w. r. t. biofertilizer?

- (1) Biofertilizers pollute soil and groundwater.
- (2) Biofertilizers increases crop yield by 15-35%.
- (3) Biofertilizers are useful in all types of soil conditions.
- (4) Biofertilizers are inexpensive



71. Among the below given biomolecules how many of them are produced by heterotropes

Citric acid, Streptomycin, Ethylalcohol, Gibberellic acid, Statins, Cyclosporine - A, streptokinase, Roquefort Cheese, Swiss Cheese

(1) 5

(2) 9

(3) 7

(4) 8



72. Match Column I with Column II

Choose the correct answer from the options given below :

Column I

A *Clostridium butylicum*

B *Saccharomyces cerevisiae*

C *Trichoderma polysporum*

D *Streptococcus* sp

Column II

P Ethanol

Q Streptokinase

R Butyric acid

S Cyclosporin-A

(1) A – R, B – P, C – Q, D – S

(3) A – R, B – P, C – S, D – Q

(2) A – Q, B – S, C – R, D – P

(4) A – S, B – P, C – R, D – Q



73. Fill up blanks by selecting the correct option

- i. ____ have been commercialized as blood cholesterol lowering agent**
- ii. ____ used as an immunosuppressive agent in organ transplant patients**
- iii. ____ is a tissue plasminogen activator**

- (1) Statins, Cyclosporin A, Streptokinase
- (2) Cyclosporin A, Statins, Chymosin
- (3) Cyclosporin A, Citric acid, Butyric acid
- (4) None of these



74. Match the columns I and II.

Column I

A LAB

B Inoculum

C Baker's yeast

Column II

P *Saccharomyces cerevisiae*

Q *Lactobacillus*

R Starter

- (1) A – R, B – P, C – Q
- (2) A – Q, B – R, C – P
- (3) A – P, B – R, C – Q
- (4) A – Q, B – P, C – R



75. Assertion : *Saccharomyces cerevisiae* produces acetic acid.

Reason : *Trichoderma polysporum* produces blood cholesterol lowering agent.

- (1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.
- (2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.
- (3) Assertion is correct but Reason is incorrect.
- (4) Both Assertion and Reason are incorrect



76. Which one of these is involved in last step of biogas production

- (1) Bacillus
- (2) Ruminococcus
- (3) Methanobacterium
- (4) Cellulomonas



77. Which of the following organism is used for prodcuton of cyclosporins?

- (1) Protist - Trichoderma polysporum
- (2) Bacteria - Trichoderma polysporum
- (3) Fungus - Trichoderma polysporum
- (4) Virus - Trichoderma polysporum



78. The biological control of agricultural pests, unlike chemical control is ____

- (1) Very expensive
- (2) Polluting
- (3) Toxic
- (4) Based on predation



79. Which one of the following antibiotics was extensively used to treat American soldiers wounded in World War-II?

- (1) Streptokinase
- (2) Statins
- (3) Penicillin
- (4) Penicillium



80. Which of the following is correctly matched for the product produced by them?

- (1) Methanobacterium : Lactic acid
- (2) Penicillium notatum: Acetic acid
- (3) Saccharomyces cerevisiae : Ethanol
- (4) Acetobacter aceti : Antibiotics



81. Match the columns I and II.

Column I

A Alexander Fleming

B Bakers yeast

C Brewers yeast

Column II

P Staphylococci

Q Aspergillus niger

R Saccharomyces cerevisiae

(1) A – R, B – P, C – Q

(2) A – Q, B – R, C – R

(3) A – P, B – R, C – R

(4) A – Q, B – P, C – R



82. The primary effluent from the ____A____ tank is passed into ____B____ tanks.

- (1) A - Settling, B - large aeration
- (2) A - Filtration, B - large aeration
- (3) A - large aeration, B - Settling
- (4) A - Settling, B - Chemical tank



83. Select the correct answer

I. Wine and rum are produced without distillation

II. Whisky, brandy and beer are produced by distillation

- (1) Only I is correct
- (2) Only II is correct
- (3) Both are correct
- (4) None are correct



84. The scientific name of Baker's yeast

- (1) *Saccharomyces cerevisiae*
- (2) *Saccharomyces ellipsoideus*
- (3) *Saccharomyces pyreformis*
- (4) *Saccharomyces ludwigii*



85. Methanogens do not produce:

- (1) Oxygen
- (2) Methane
- (3) Hydrogen sulfide
- (4) Carbon dioxide



86. Statement A : Ladybird and aphids are used as biopesticide

Statement B : Biopesticides control weeds and pest without causing any damage to living organisms.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



87. BOD of waste water is estimated by measuring the amount of:

- (1) Total organic matter
- (2) Biodegradable organic matter
- (3) Oxygen evolution
- (4) Oxygen consumption.



88. Statement A : Nowadays people realised that there is a large pressure to switch to organic farming.

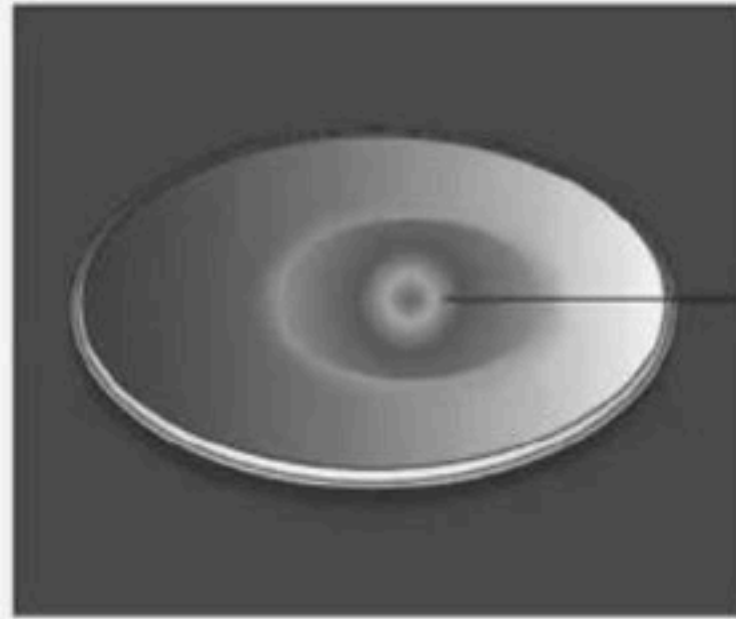
Statement B : Biofertilizers are organisms like cyanobacteria that enrich the nutrient quality of soil.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



89. Which microorganism is growing on potato dextrose agar containing petri plate?

- (1) Bacteria
- (2) Fungi
- (3) Virus
- (4) Virioids



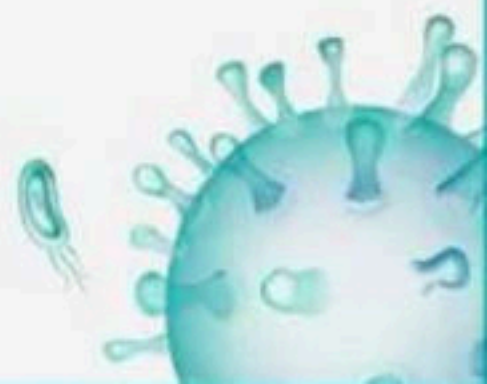
90. All solids that settle form the ____i____, and the ____ii____ forms the effluent.

- (1) i- secondary sludge, ii- supernatant
- (2) i- primary sludge, ii- supernatant
- (3) i- primary sludge, ii- settled matter
- (4) i- secondary sludge, ii- settled matter



91. Which is the correct statement on a good antibiotic?

- (1) A good antibiotic is the one which is harmless to host its microflora but is effective against all strains of pathogens.
- (2) A good antibiotic is the one which is harmful to host and its microflora
- (3) A good antibiotic should not be effective against all strains of pathogen
- (4) None of these



92. An important part of the biological farming approach is

i. To become familiar with the various life forms that inhabit the field

ii. Predators as well as pests, and also their life cycles

iii. Patterns of feeding and the habitats that they prefer.

(1) Only i, ii

(2) Only i, iii

(3) Only ii, iii

(4) All of them



93. Match the columns I and II.

Column I		Column II	
A	<i>Trichoderma</i>	P	BGA
B	<i>Nostoc</i>	Q	Fungi
C	Baculovirus	R	NPV

- (1) A – R, B – P, C – Q,
- (2) A – Q, B – R, C – P,
- (3) A – P, B – R, C – Q,
- (4) A – Q, B – P, C – R



94. Primary treatment of wastes is the

- (1) Physical removal of large and small particles from sewage
- (2) Biological removal of large and small particles from sewage
- (3) Both (1) and (2)
- (4) Chemical removal of large and small particles from sewage



95. As the organic matter increases in a water body, the

BOD:

- (1) Increases
- (2) Decreases
- (3) Remains unchanged
- (4) Not a parameter



96. Which one of the following is/ are excellent candidates for species specific, narrow spectrum insecticidal applications?

- (1) Virus
- (2) Bacteria
- (3) Virioids
- (4) Prions



97. Cyclosporin A, an immunosuppressive agent commercially obtained from

- (1) *Trichoderma viridea*
- (2) *Trichoderma harzianum*
- (3) *Trichoderma polysporum*
- (4) *Trichoderma reesei*



98. Match the microbial products listed under Column I with the related microbes given under Column II; choose the appropriate option from the given choices

Column I		Column II	
A	Citric acid	P	<i>Methanobacterium</i>
B	Cyclosporin A	Q	<i>Monascus purpureus</i>
C	Statin	R	<i>Aspergillus niger</i>
D	Gobar gas	S	<i>Trichoderma polysporum</i>
		T	<i>Clostridium butylicum</i>

- (1) A - R, B - S, C - Q, D - P
- (2) A - T, B - Q, C - S, D - R
- (3) A - Q, B - S, C - T, D - R
- (4) A - R, B - S, C - Q, D - T



99. *Saccharomyces cerevisiae* is employed in making

- (1) Idli
- (2) Beer
- (3) Bread
- (4) 2 and 3



100. Formation of flocs means

- (1) Association of virus with the bacterial remains in water
- (2) Bacteria which is associated with the mycorrhiza of the fungus of soil
- (3) Bacteria which is associated with the mycorrhiza of the fungus of water
- (4) Bacteria which is associated with the fungi in solid waste of the water



THANK YOU



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